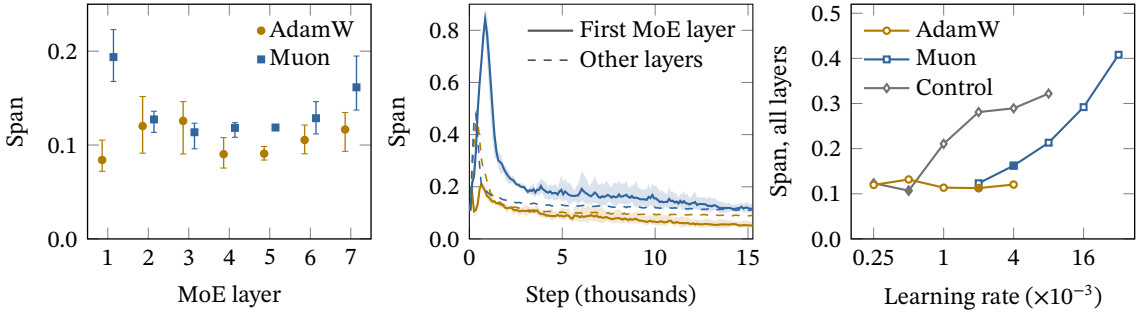

AdamW
Muon
Span
0.0
0.1
0.2
1
2
3
4
5
6
7
MoE layer
First MoE layer
Other layers
Span
0.0
0.2
0.4
0.6
0.8
0
5
10
15
Step (thousands)
AdamW
Muon
Control
Span, all layers
0.0
0.1
0.2
0.3
0.4
0.5
0.25
1
4
16
Learning rate (×10⁻³)